

Hey, everybody. Here we are at Chapter 23 review. Believe it or not, this is the final episode in our series of chats. And it's a good one to wrap things up with because we get to focus on the reason a lot of us got into the fitness industry-- helping people. The information presented up to this point has focused primarily on exercise considerations for apparently healthy adults. This chapter covers important information and recommendations for assessing and designing exercise programs for clients with chronic health conditions or functional limitations.

The fact of the matter is, these types of clients are a growing segment of society. Roughly a third of the world's population is categorized as overweight or obese. And many more individuals deal with at least one chronic health condition every day. Age also impacts training decisions, as do other normal health scenarios and life changes, like pregnancy.

Research has shown that progressive exercise and increased physical activity can significantly improve the health and well-being of all sorts of special populations. To serve the widest available clientele, it's important for you to understand the intricacies of program design for a variety of health variables. Fitness professionals are encouraged to continue to learn and expand their knowledge to keep pace with the rapidly growing field of clinical exercise physiology.

At the end of this chapter, you should be able to identify exercise needs and limitations for different populations, summarize the effects of chronic disease, health limitations, and challenges on clients' abilities to respond and adapt to exercise, differentiate exercise considerations for clients who have a chronic disease, determine risk factors for chronic disease and other physical or health challenges, as well as identify considerations taken to accommodate special populations when exercising.

When focusing on client care and service, it's important to understand there are fundamental physiologic differences between individuals who are apparently healthy and those with special circumstances. Although clients with special conditions may experience similar effects from exercise training as apparently healthy individuals, some don't respond, adapt, or progress in the same way.

As a CPT, it's imperative to be aware of and sensitive to these variables. Let's take a look at each of these special populations, starting with age considerations. When we think of kids in a fitness program, it's usually relative to whatever sport they're playing. However, there's a significant need for increased awareness and access to year-round general fitness training for the youth population outside of the sports seasons.

Childhood obesity and diabetes has been rapidly increasing, and youth fitness guidelines are needed to focus on promoting healthy lifestyles and health-related physical fitness on a consistent basis. If you're dealing with young athletic clients, you'll likely see something interesting-- the proliferation of early specialization. That can take a toll on the body as well. Even young athletes can benefit from variability in their exercise and movement programming as it relates to their longterm musculoskeletal health.

So don't think just because a kid is super sporty already that a structured fitness program won't be of use to them. In general, getting kids on a program early on is going to set them up with healthy habits for life. It's a strategy that can prevent a lot of problems down the road. Current guidelines recommend that kids get an hour of moderate to high-intensity activity each day. And research has shown that it's safe and effective to use resistance training for youth clientele.

There are fundamental physiological differences between children and adults to be aware of. How they respond to exercise is different. And therefore, how the exercise program is developed and implemented should also be different. Now, at the opposite end of the spectrum, you'll also work with an aging population. Despite the normal decline in physiologic functioning associated with aging, older adults, both with and without other chronic health conditions, can often respond to exercise similarly to apparently healthy younger adults.

By sticking with a systematic process, you can make a dramatic impact on the overall health and well-being of older adults. Bear in mind, however, that normal and abnormal pathologic changes associated with aging affect the response to exercise training. Degenerative processes associated with aging can lead to a decrease in the functional capacity of older adults, including potentially significant declines in muscular strength and endurance, range of motion, mobility, cardiorespiratory fitness, and proprioceptive neural responses.

Before initiating any exercise training, older adults must complete a PAR-Q+ and other appropriate assessments. This will allow you to match exercises to your clients' capabilities in a safe and effective manner. Ensure all exercises are performed with precise technique and control to minimize the risk of injury, and consult with a licensed physician whenever questions arise.

Let's switch gears now a little bit to discuss chronic health conditions. First, the fastest-growing health problem worldwide is obesity. It's a disease that leads to many other health conditions as well as emotional and social issues. Although the causes are complex and varied, few disagree the primary problem is energy balance. Regular physical activity and exercise to expend calories is one of the most important factors related to longterm successful weight loss.

Exercise has also been shown to have a substantial positive effect on the treatment and prevention of type 2 diabetes, which can be related to obesity, and type 1 diabetes, in which the body does not produce enough insulin. The most important goals of exercise for individuals with either type are glucose control, weight loss, and energy expenditure. Be aware of strategies to prevent hypoglycemic-- low blood glucose-- and hypoglycemic-- elevated blood sugar events-- as well as when to defer exercise based on resting blood glucose levels.

Hypertension, or high blood pressure, is another common medical disorder. Medication is a common method of controlling this. Although, comprehensive lifestyle changes, including regular fitness activity, have also been shown to reduce blood pressure. When it comes to matters of the heart, exercise can be a huge benefit. For clients with stable coronary artery disease, for example, it can lower the risk of mortality, increase exercise tolerance and muscle strength, reduce angina and heart failure symptoms, as well as improve physiological status and social adjustment.

Other common areas of concern are bones and joints. For clients with osteopenia and osteoporosis, exercise programs that combine resistance training with flexibility, core, and balance training increases bone mass density and enhances proprioception. In terms of arthritis, it's important to understand the difference between rheumatoid arthritis and osteoarthritis and be aware of the signs and symptoms of an acute rheumatoid arthritis flare-up. You'll also need to monitor the progress of clients with arthritis to assess the effects of the exercise program on joint pain.

A condition we may not immediately connect to exercise benefits is cancer, but it's actually hugely important to aid in recovery. It can improve exercise tolerance, reduce the cellular risks associated with cancer, and improve quality of life, both physically and mentally. Core and balance exercises are essential for this population, as they will help in regaining the stabilization necessary for activities of daily living that may have been compromised due to a lack of activity caused by treatments. Clients should be progressed slowly and according to their physical ability.

Pregnancy is another interesting area of fitness study. There's been substantial research documenting the benefits of exercise on the physiology and health of both the mother and developing fetus. Most recreational pursuits are appropriate for all pregnant women until the third trimester, when a logical reduction in activity is recommended. During the postpartum period, there may be a tendency to rush into and through an exercise program in an effort to return to pre-pregnancy fitness and body composition. But it'll be your job to advise these clients to take it slow.

Chronic lung disease is also a common health condition. While smoking has progressively declined over the last few decades, it is still one of the leading preventable causes of death and a primary risk factor for the development of chronic lung disease. This can be largely broken into two major categories, obstructive and restrictive. However, the impairments during exercise are similar with both. Problems include decreased ventilation and gas exchange ability, which results in decreased aerobic capacity and oxygen desaturation.

Clients with lung disease experience fatigue at low levels of exercise and often have dyspnea or shortness of breath. Exercise can help improve functional capacity and decrease the symptoms of dyspnea, among many other physiologic and psychological benefits. Finally, we have Peripheral Arterial Disease, or PAD. It's defined as the narrowing of major arteries that are responsible for supplying blood to the lower extremities, characterized by limping, lameness, or leg pain.

As a fitness professional, your job will be to develop a training regimen that attempts to improve physical function in the face of limiting factors. PAD is also associated with coronary heart disease and diabetes. So be aware of other existing or undiagnosed comorbidities. Consult a physician with any concerns.

For every situation, you should focus on educating your clients about the benefits associated with exercise and tailoring their programs in a mindful way to achieve success within their own special needs and concerns. You'll find the upsides will be similar for a lot of different circumstances. But the details and specific benefits and contraindications will vary from person to person and from condition to condition. You'll need to be aware of the recommended modifications in terms of exercise selection, volume, intensity, time, modalities, and so forth to be sure you're supporting your clients during every stage of life, no matter what health issues they're facing.

Like I said at the beginning, it's always beneficial to do additional research, particularly if you end up working with a client within one of these groups. Who knows, you might even decide that you want to work exclusively with a particular population and specialize in programming for their unique needs. And with that, I am bringing our series of discussions to a close. We have completed each chapter in this program, but you have just begun.

The simple fact that you are actively studying and taking advantage of all the great resources NASM has available speaks volumes about your commitment and passion. Best of luck to you with the rest of your studying, training, certification exams, and launching your career as a CPT. Take care, and be well.